

Experiment 5: Implementation of Docker Commands – Complete Summary

This is a good viva revision sheet for you.

1. What is Docker?

Docker is a containerization platform used to create, run, and manage containers.

Why Docker?

- **Consistent environment**
 - **Easy deployment**
 - **Lightweight**
 - **Supports DevOps and CI/CD**
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2. Installation of Docker

Download Docker Desktop

Install Docker Desktop and start it.

Verify Installation

`docker --version`

Output:

Docker version 29.5.2

This confirms Docker is installed.

3. Verify Docker is Working

Run:

`docker run hello-world`

Docker:

1. **Downloads hello-world image**
2. **Creates container**
3. **Runs container**
4. **Displays message**
5. **Stops container**

Output:

Hello from Docker!

This confirms Docker daemon is working.

4. What is an Image?

An Image is a template used to create containers.

Real-Life Example

House Blueprint = Image

Built House = Container

Image contains:

- **Operating System**
 - **Libraries**
 - **Dependencies**
 - **Application Code**
-

5. Download an Image

We downloaded Ubuntu image.

docker pull ubuntu

Docker downloaded Ubuntu from Docker Hub.

6. View Images

docker images

Output:

hello-world

ubuntu

These are images stored on your PC.

7. What is a Container?

A Container is a running instance of an image.

Flow

Image



Container



Running Application

8. Create a Container

Command:

docker run --name mycontainer -it ubuntu bash

Meaning:

Option

Meaning

docker run

Create and run container

Option	Meaning
--name mycontainer	Name of container
-it	Interactive terminal
ubuntu	Image name
bash	Open Linux shell

9. Enter Ubuntu Container

After running:

docker run --name mycontainer -it ubuntu bash

Prompt changes to:

root@xxxx:/#

Meaning:

You are inside Ubuntu Linux

even though host OS is Windows.

10. Check Ubuntu Version

Inside container:

cat /etc/os-release

Displays Ubuntu details.

11. Exit Container

exit

Container stops.

Status becomes:

Exited (0)

12. View Running Containers

docker ps

Shows only running containers.

Example:

CONTAINER ID

IMAGE

STATUS

13. View All Containers

docker ps -a

Shows:

- **Running containers**
- **Stopped containers**

You saw:

mycontainer
hello-world

14. Start a Stopped Container

docker start mycontainer

Container starts again.

15. Stop a Running Container

docker stop mycontainer

Container stops.

16. Remove a Container

docker rm mycontainer

Container gets deleted.

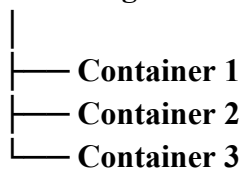
17. Remove an Image

docker rmi ubuntu

Deletes Ubuntu image.

18. Relationship Between Image and Container

Ubuntu Image



One image can create multiple containers.

Docker Commands Used

Command

docker --version

docker run hello-world

Purpose

Check Docker version

Verify Docker

Command	Purpose
docker pull ubuntu	Download image
docker images	List images
docker run --name mycontainer -it ubuntu bash	Create container
docker ps	Show running containers
docker ps -a	Show all containers
docker start mycontainer	Start container
docker stop mycontainer	Stop container
docker rm mycontainer	Remove container
docker rmi ubuntu	Remove image

Important Viva Questions

What is Docker?

Docker is a containerization platform used to create, run, and manage containers.

What is an Image?

A read-only template used to create containers.

What is a Container?

A running instance of a Docker image.

Difference Between Image and Container?

Image	Container
Template	Running Instance
Read-only	Executable
Creates Containers	Created From Image

Command to View Images?

docker images

Command to View Running Containers?

docker ps

Command to View All Containers?

docker ps -a

Command to Stop Container?

docker stop mycontainer

Command to Start Container?

docker start mycontainer

One-Line Experiment Conclusion

Successfully installed Docker, pulled Ubuntu image, created and managed containers using Docker commands, and understood the concepts of Images and Containers.

 **At this point, you have completed and understood Experiment 5: Implementation of Docker Commands. The next experiment (Experiment 6) will use these concepts to create your own Docker image and container using a Dockerfile.**